

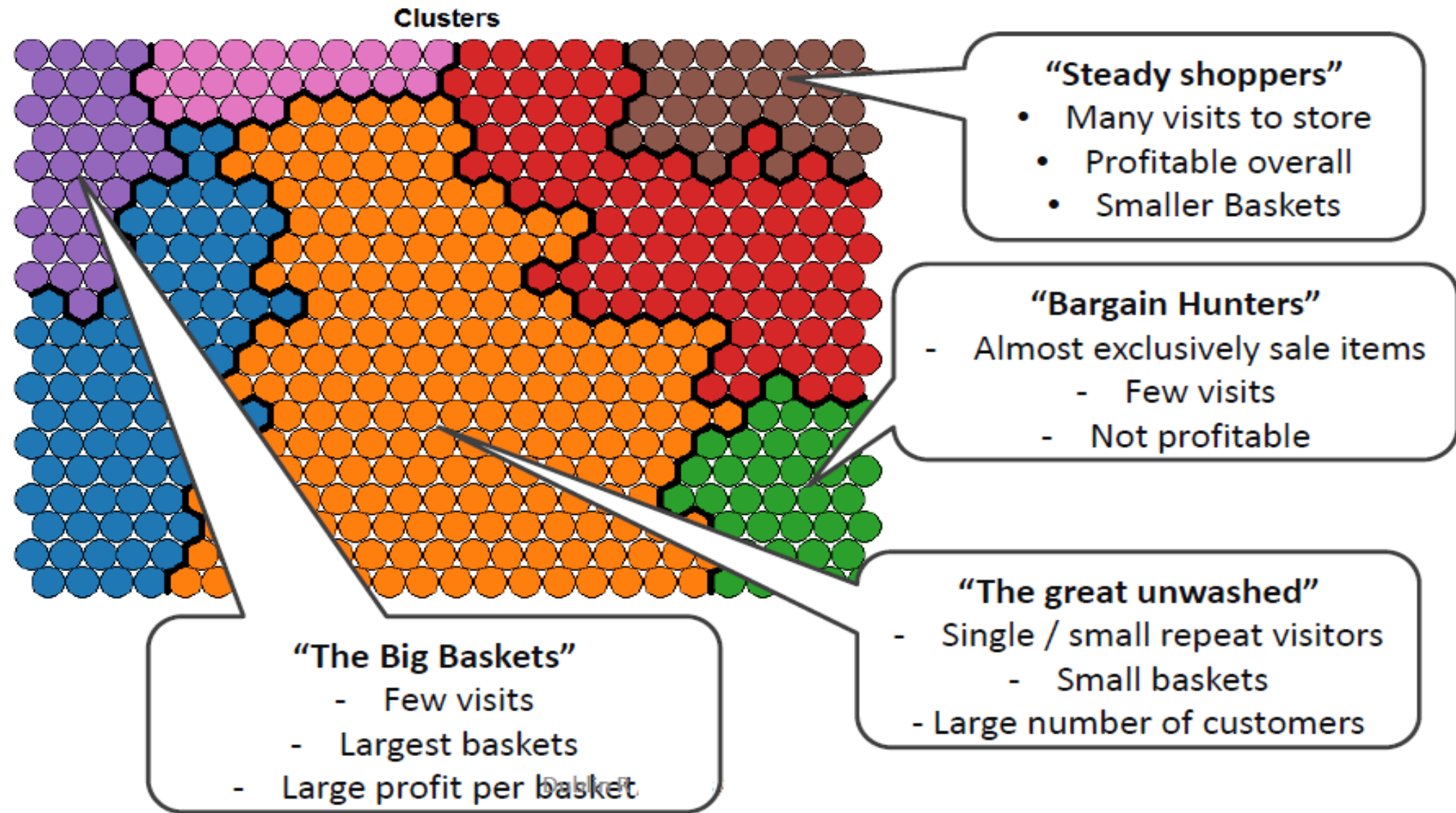
Clustering



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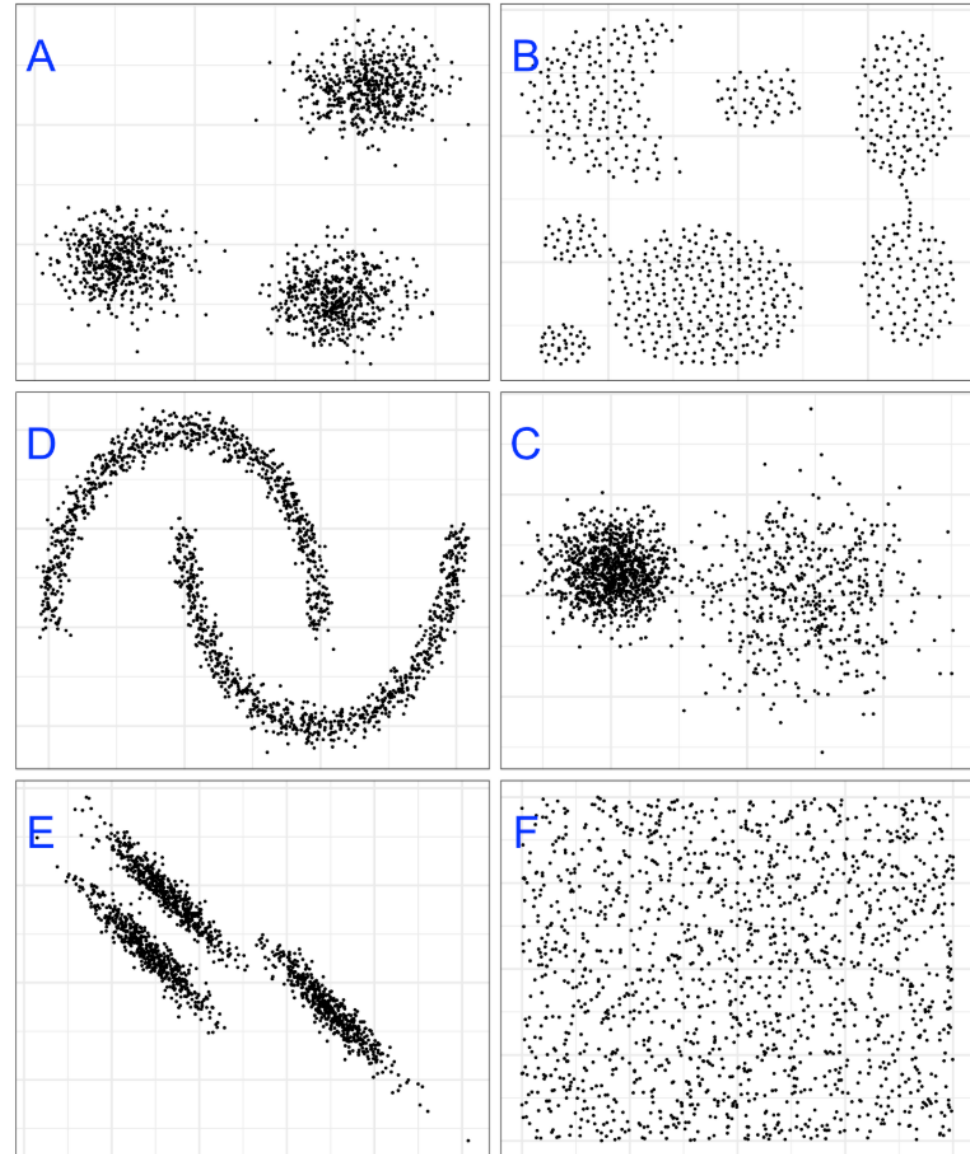
Unsupervised Learning

Clustering



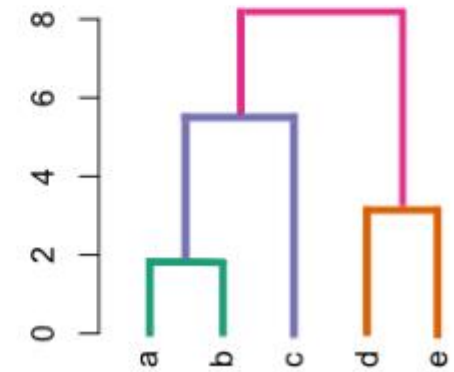
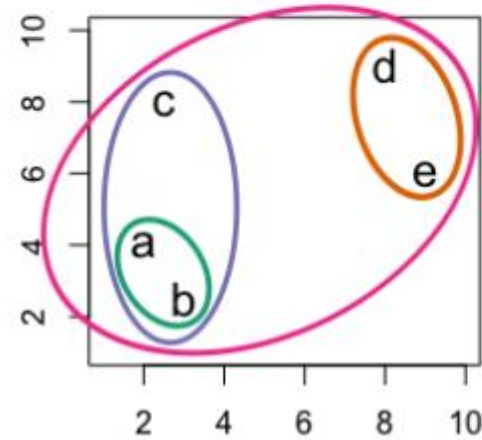
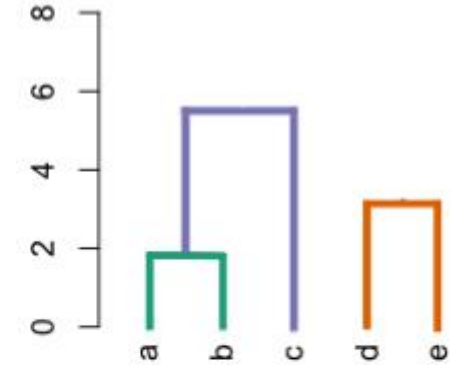
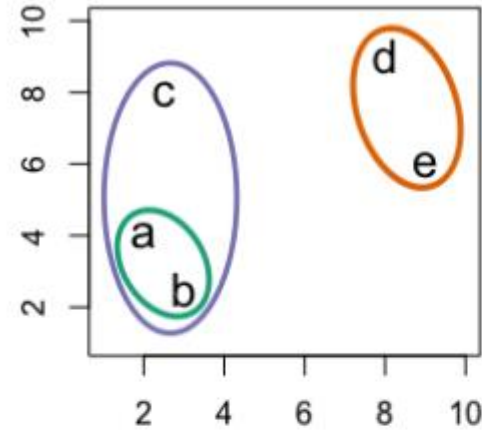
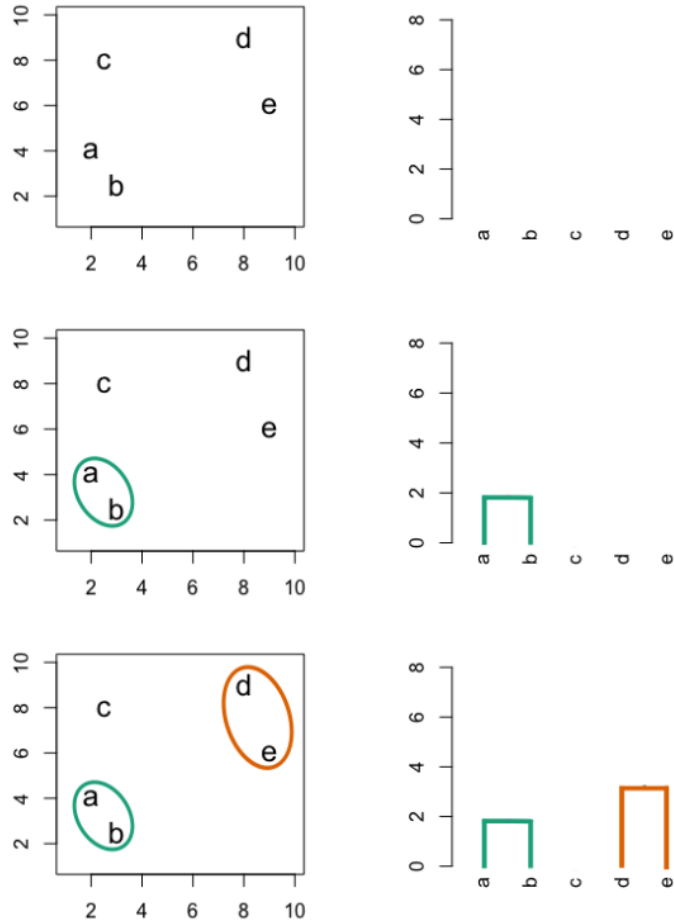
Clustering

- Clustering attempts to find groups (clusters) of similar objects. The members of a cluster should be more similar to each other, than to objects in other clusters. Clustering algorithms aim to minimize intra-cluster variation and maximize inter-cluster variation.
- Methods of clustering can be broadly divided into two types:
- Hierarchic techniques produce dendrograms (trees) through a process of division or agglomeration.
- Partitioning algorithms divide objects into non-overlapping subsets (examples include k-means and DBSCAN)



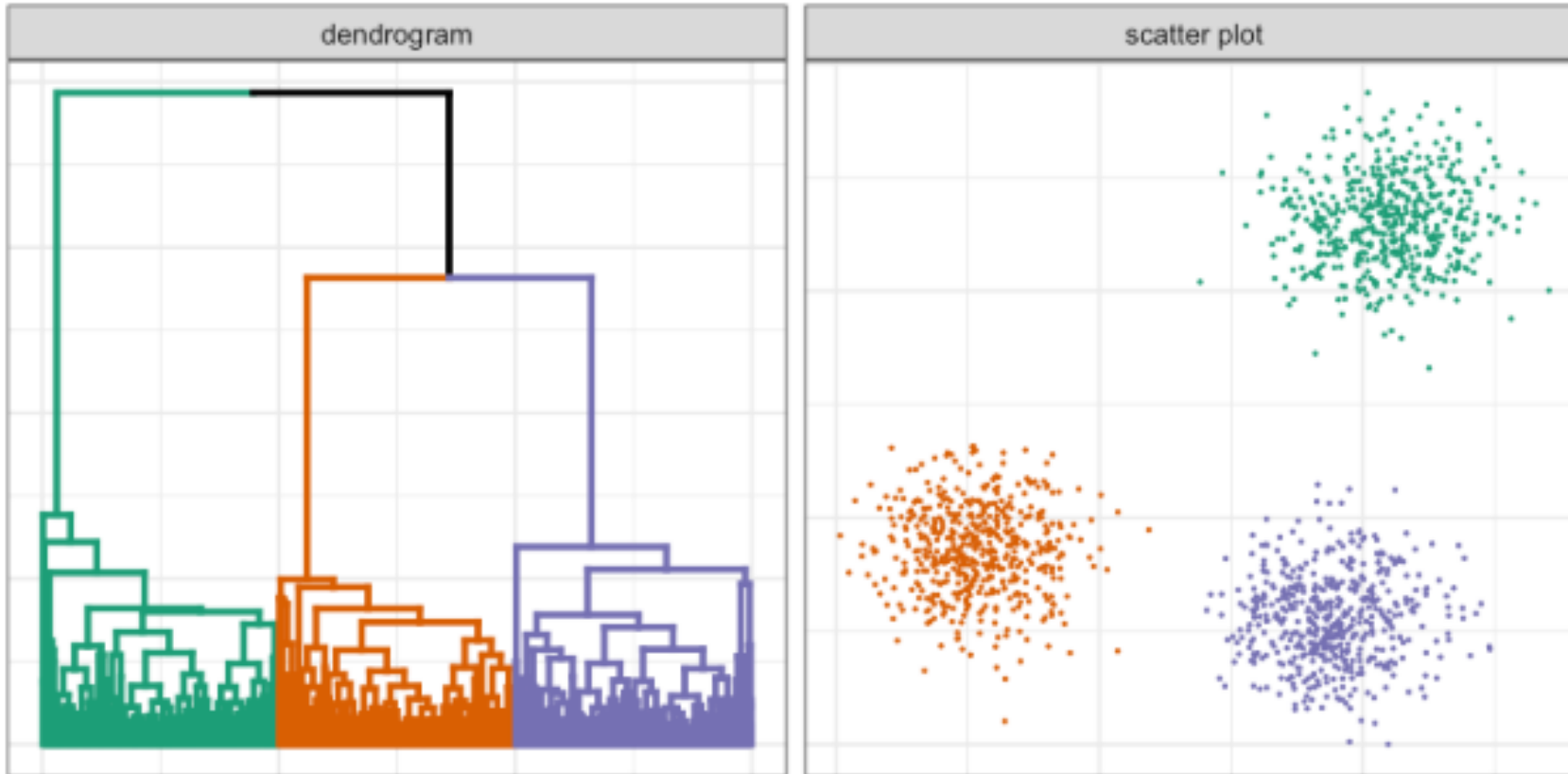
K-Means

Hierarchic agglomerative



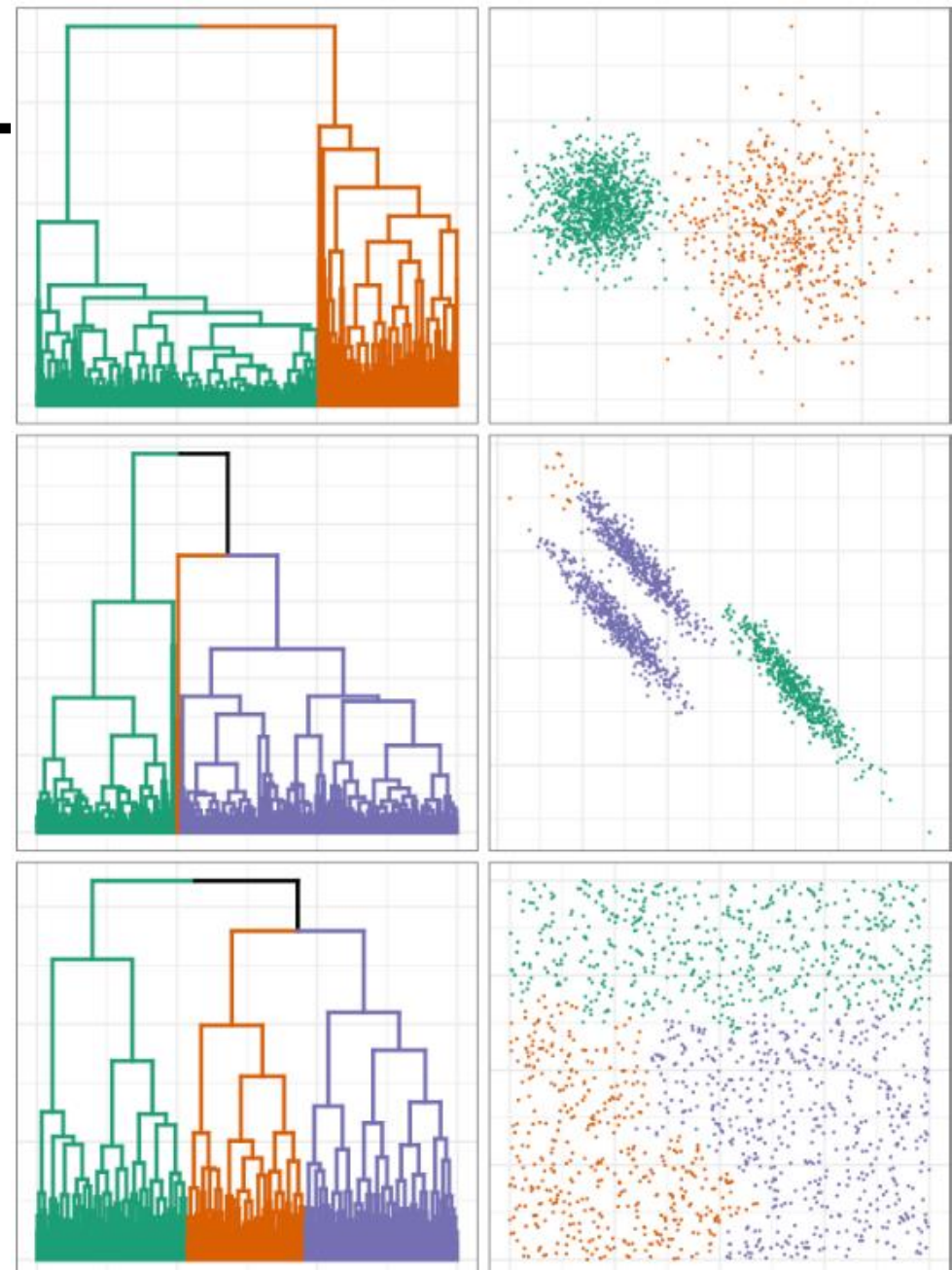
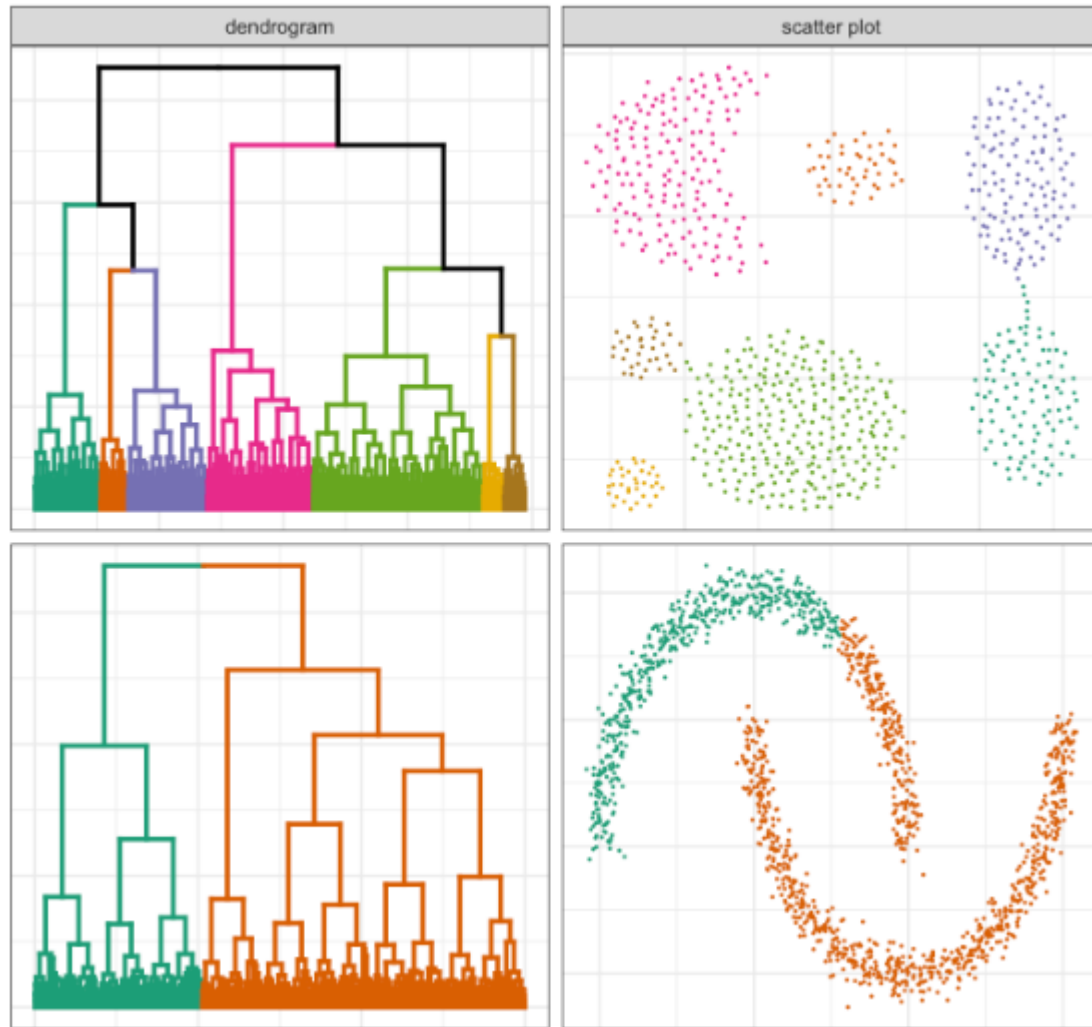
K-Means

Hierarchic agglomerative

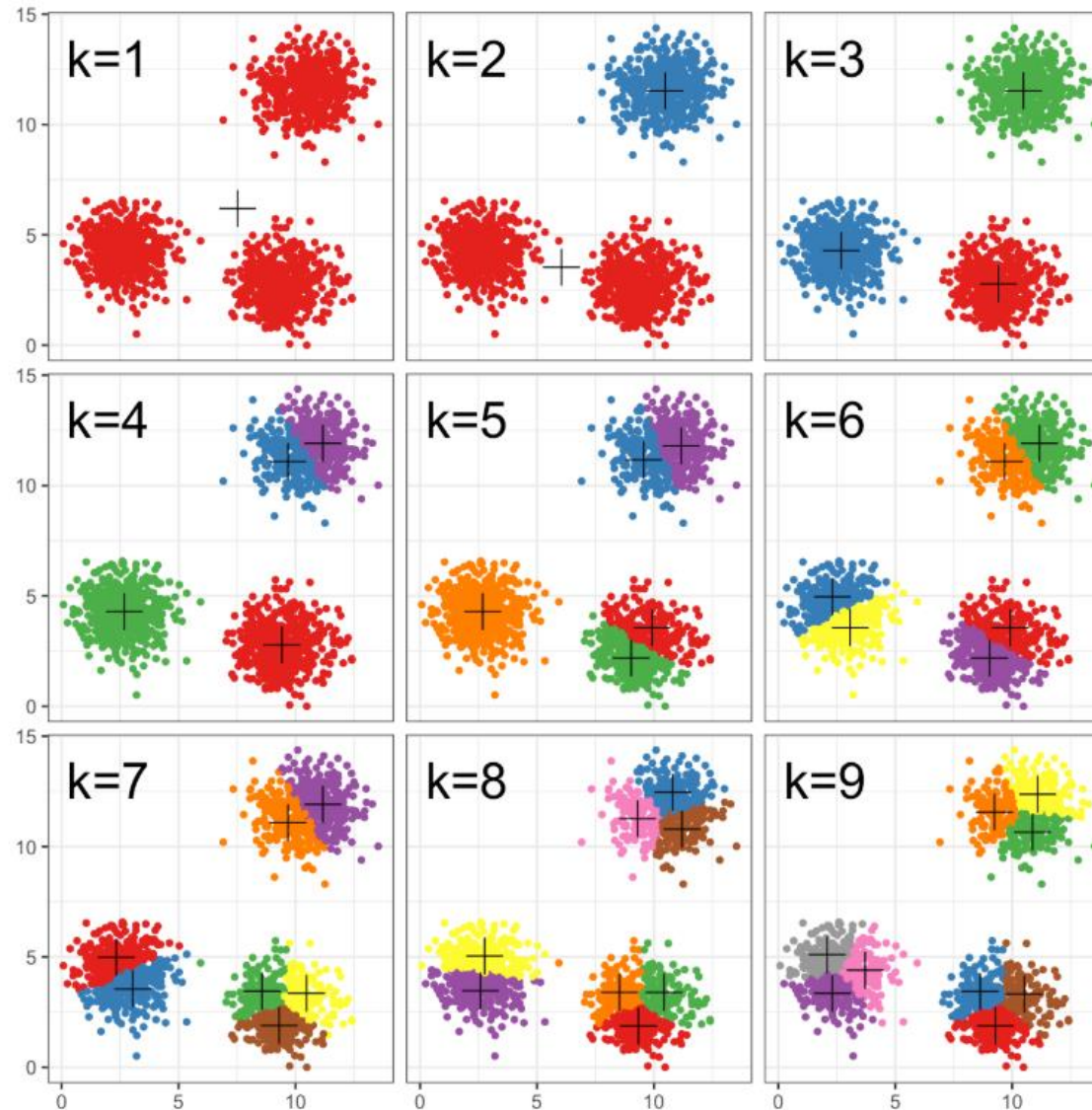


K-Means

Hierarchic agglomerative

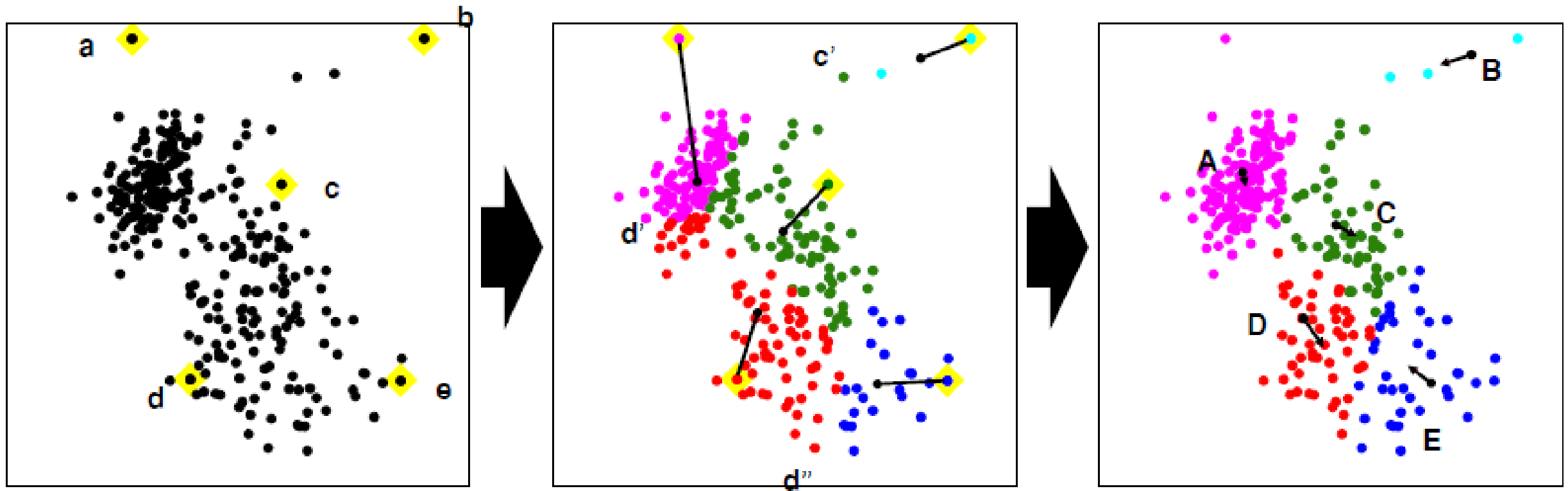


K-Means



K-Means

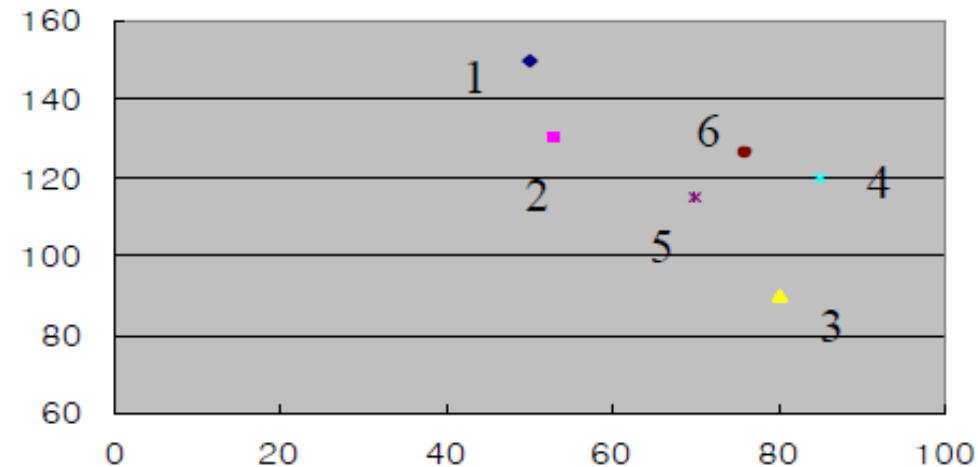
Algorithm



K-Means

K-means Clustering Example

ID	X1	X2
1	150	50
2	130	53
3	90	80
4	120	85
5	115	70
6	126	76



각 개체간 거리

ID	1	2	3	4	5	6
1	0.0					
2	20.2	0.0				
3	67.1	48.3	0.0			
4	46.1	33.5	30.4	0.0		
5	40.3	22.7	26.9	15.8	0.0	
6	35.4	23.3	36.2	10.8	12.5	0.0

K-Means

K-means Clustering Example

● K=2

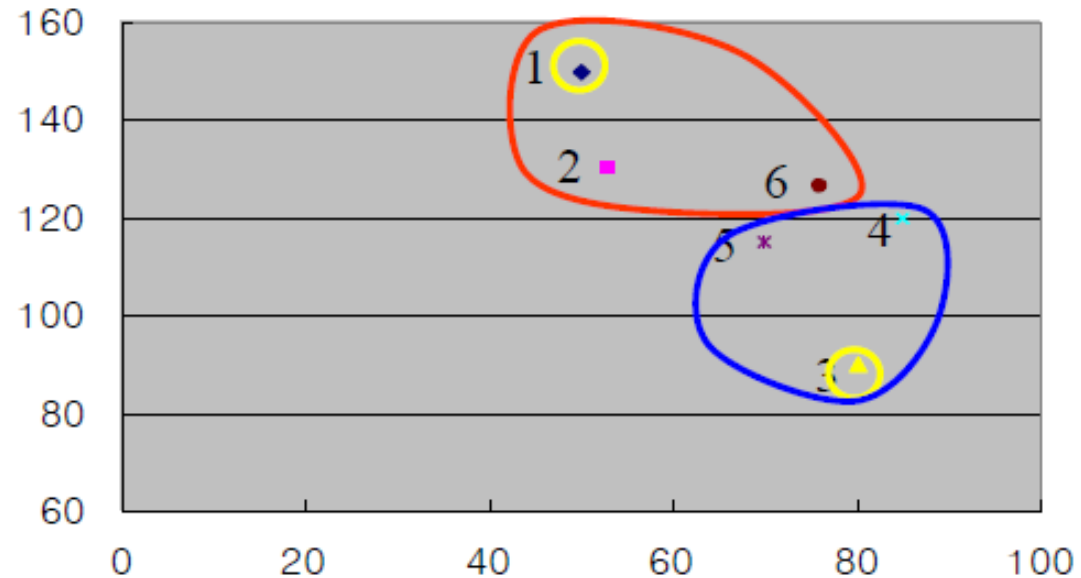
Iteration 1

Step 0

Seeds (초기중심)

ID	1	3
1	0.0	67.1
2	20.2	48.3
3	67.1	0.0
4	46.1	30.4
5	40.3	26.9
6	35.4	36.2

Step 1



K-Means

K-means Clustering Example

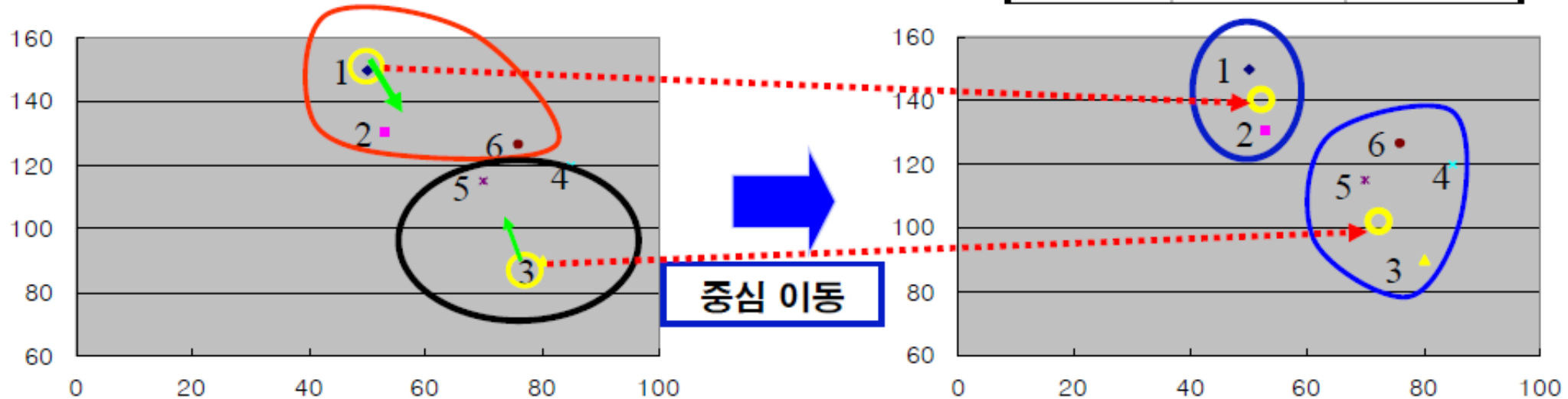
Step 2

	군집1	군집2
개체	1, 2, 6	3, 4, 5
군집중심	(59.6, 135.3)	(78.3, 108.3)

→ centroids

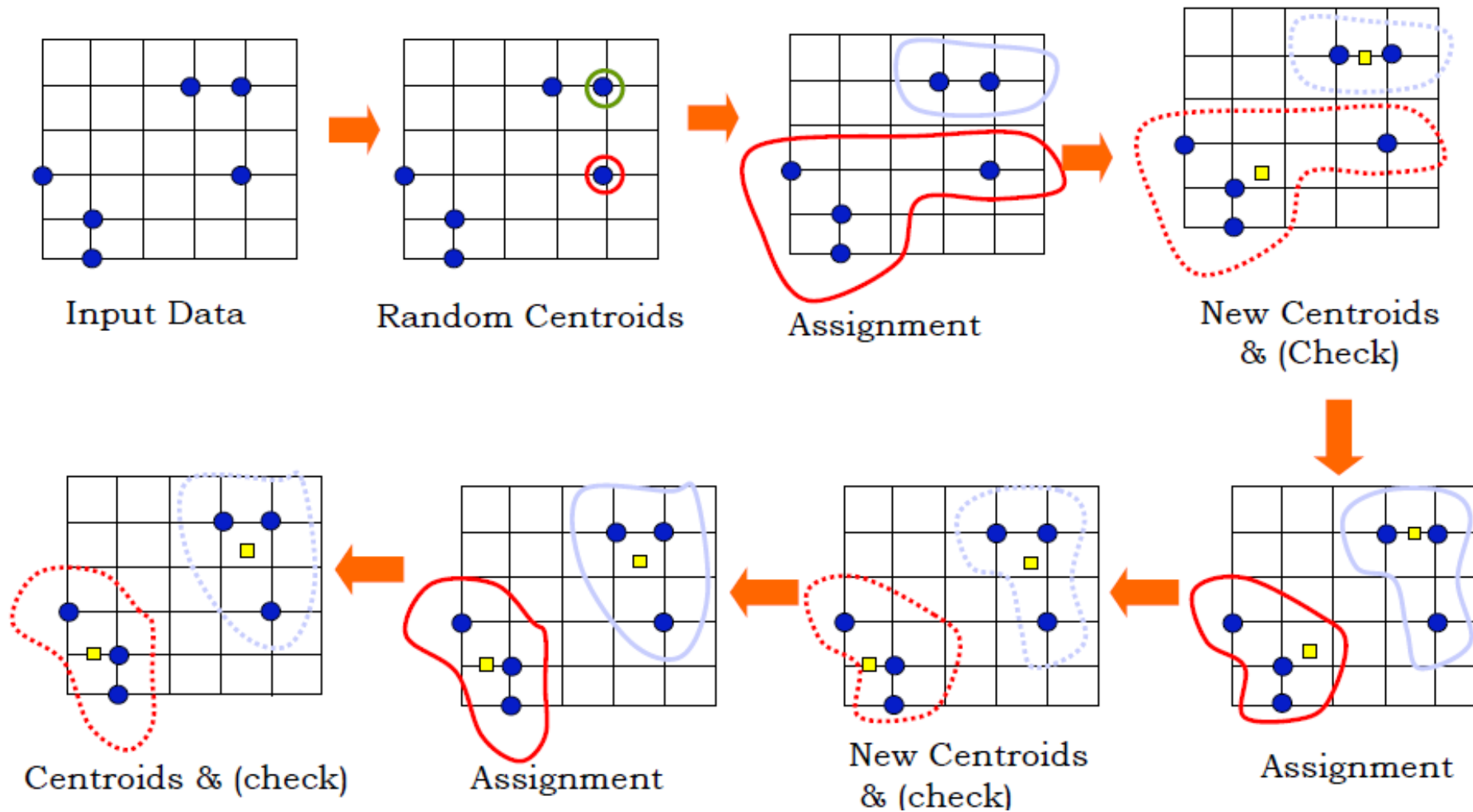
Iteration 2

	군집1	군집2
1	17.56	50.40
2	8.46	33.33
3	49.68	18.38
4	29.65	13.48
5	22.81	10.67
6	18.85	17.85



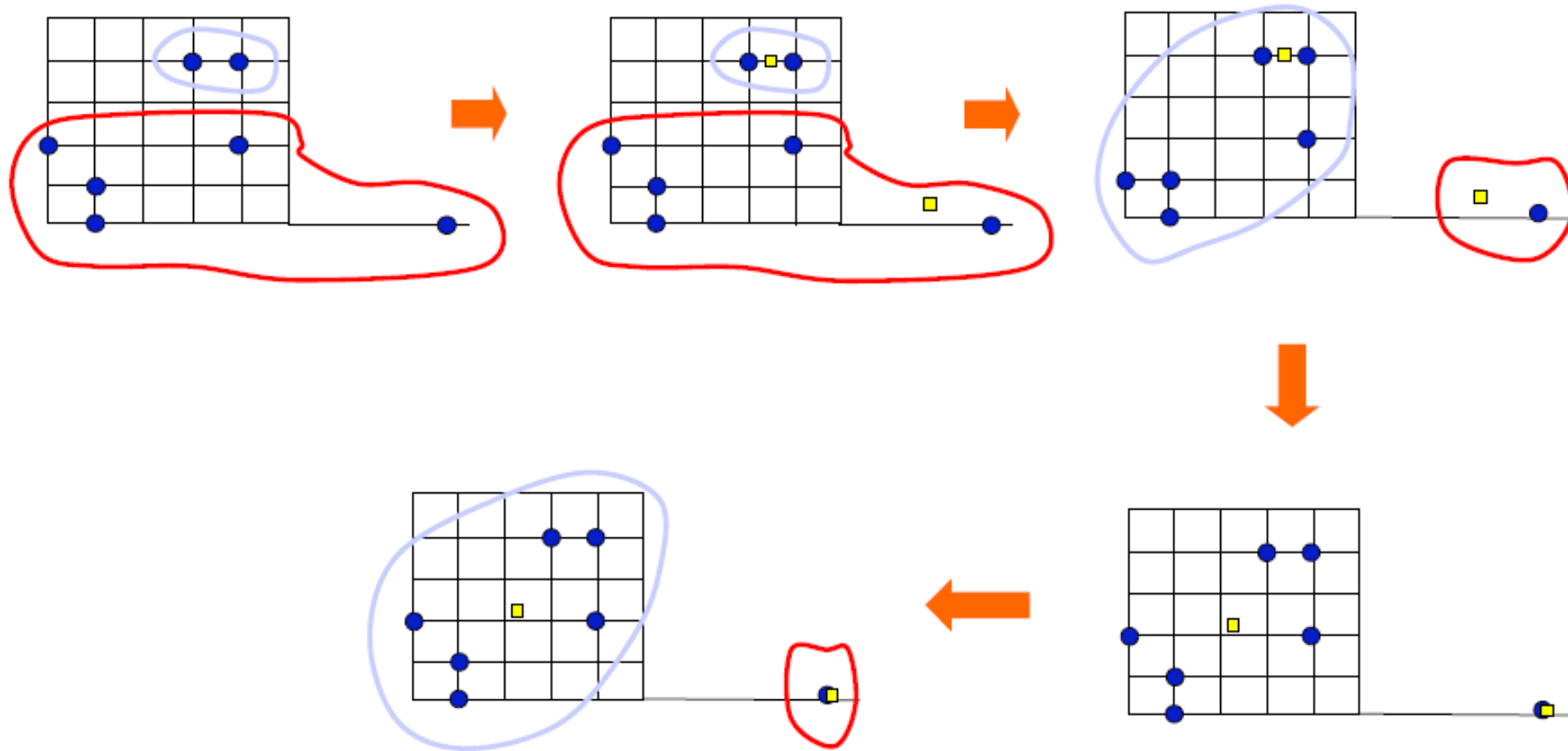
K-Means

❖ simple example



K-Means

❖ weakness on outlier (noise)



K-Means

K-means Clustering Example

● K=2

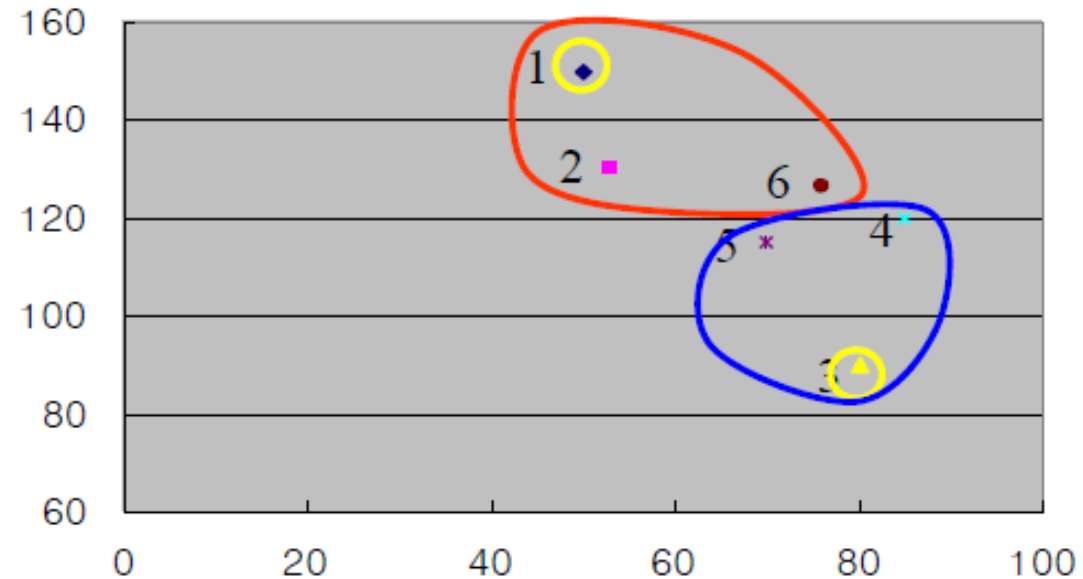
Iteration 1

Step 0

Seeds (초기중심)

ID	1	3
1	0.0	67.1
2	20.2	48.3
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Step 1



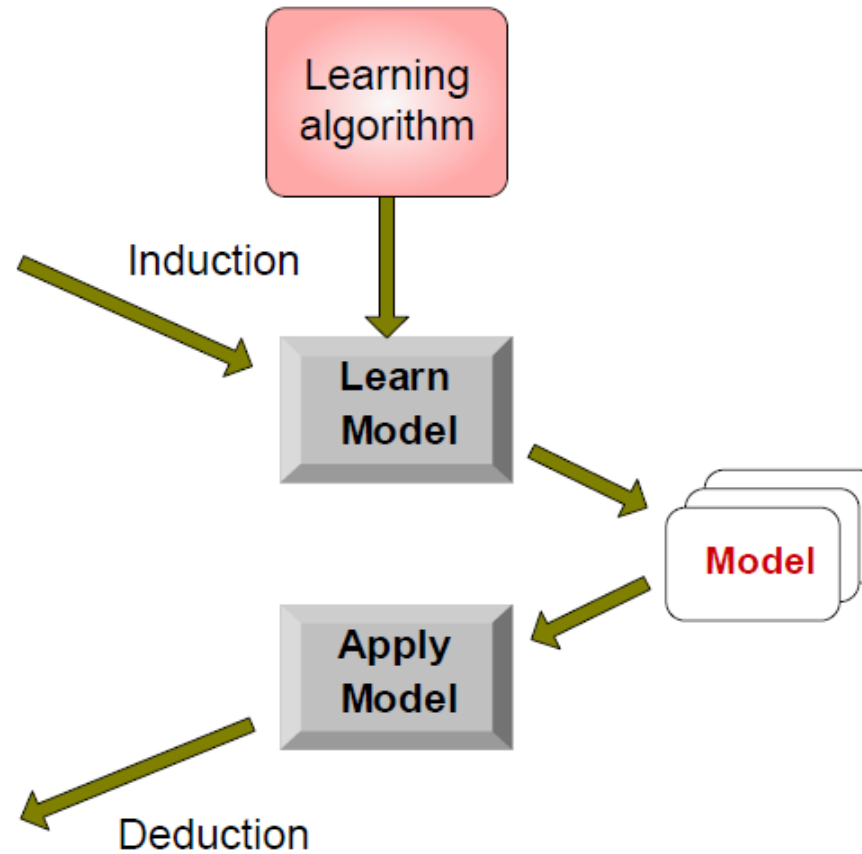
Classification

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

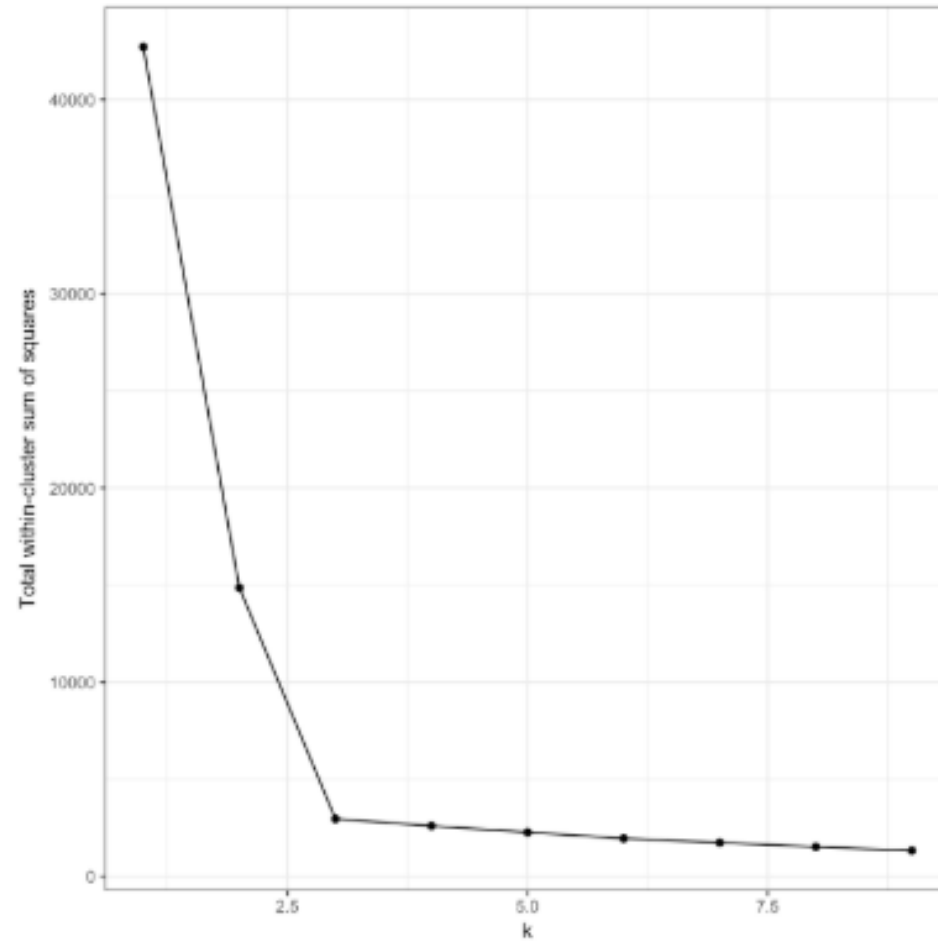
Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Classification



Classification

Silhouette

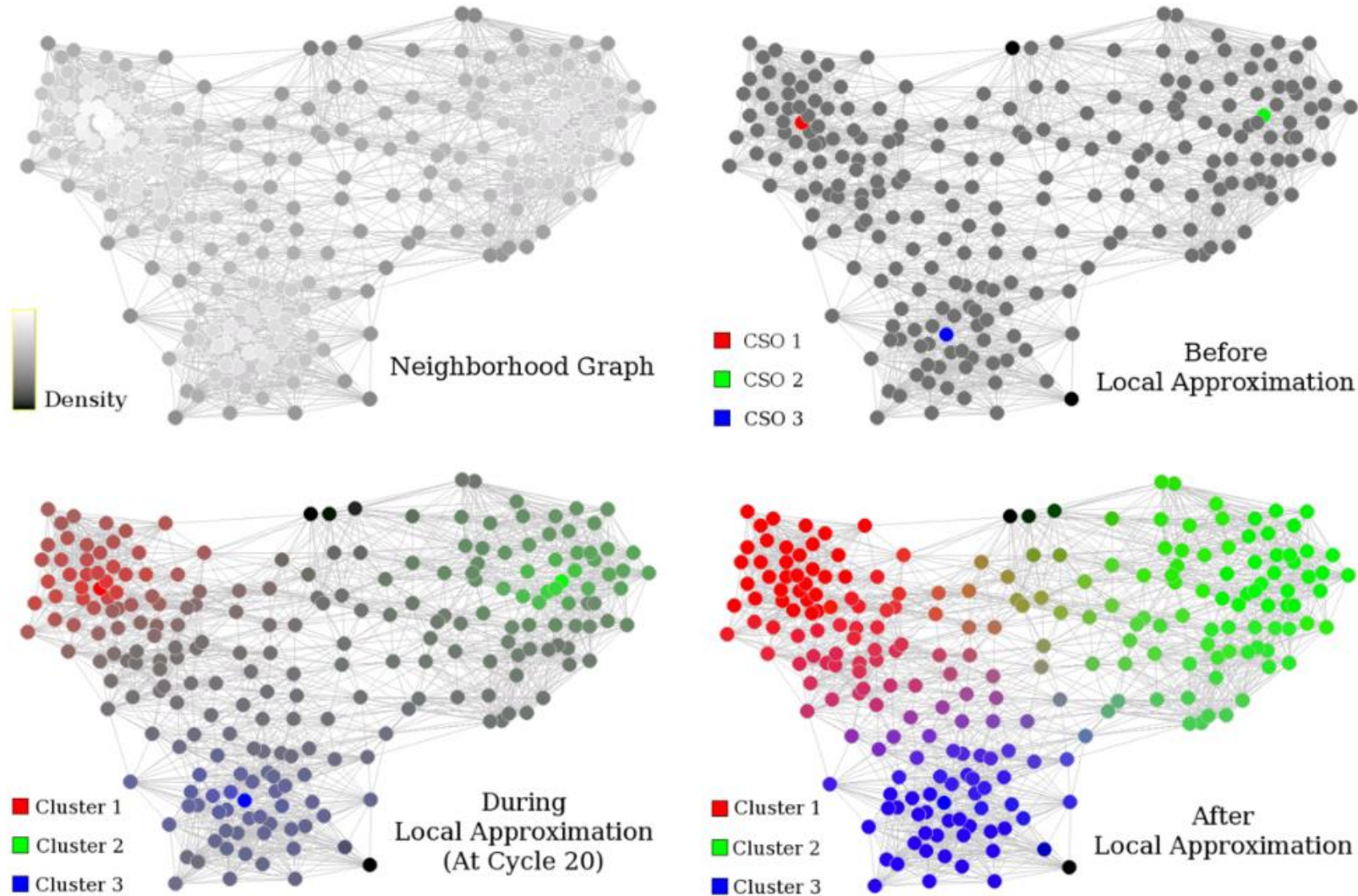
$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (10.1)$$

Where

- $a(i)$ - average dissimilarity of i with all other data within the cluster. $a(i)$ can be interpreted as how well i is assigned to its cluster (the smaller the value, the better the assignment).
- $b(i)$ - the lowest average dissimilarity of i to any other cluster, of which i is not a member.

Observations with a large $s(i)$ (close to 1) are very well clustered. Observations lying between clusters will have a small $s(i)$ (close to 0). If an observation has a negative $s(i)$, it has probably been placed in the wrong cluster.

Classification



Question and Answer
THANKYOU 😊